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Intro to Electrical Engineering

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Lab Report #5

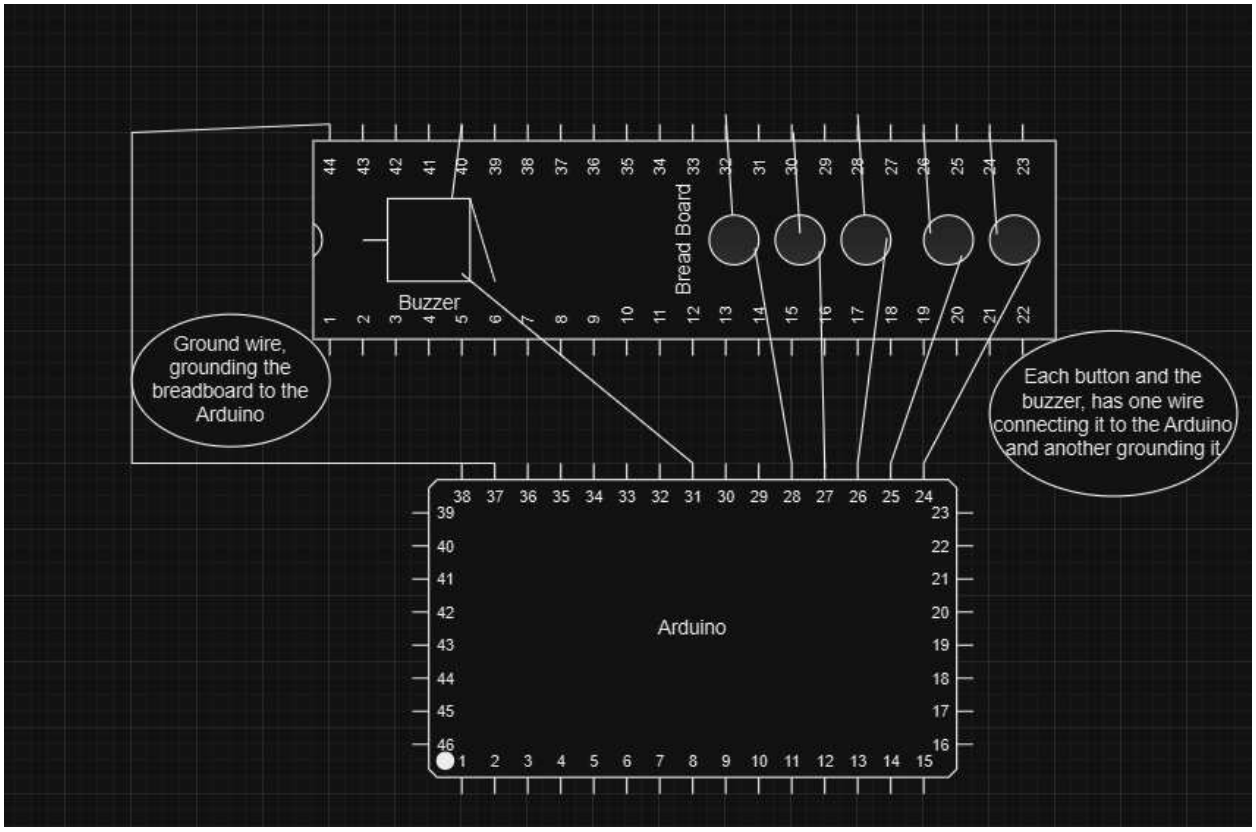
Overview

On the breadboard, we are going to wire five buttons and one buzzer and make a keyboard. So each button click is going to make the buzzer activate and each click is going to play the buzzer at a different frequency. This is going to make a piano and allow us to play sounds at different tones just like a normal piano.

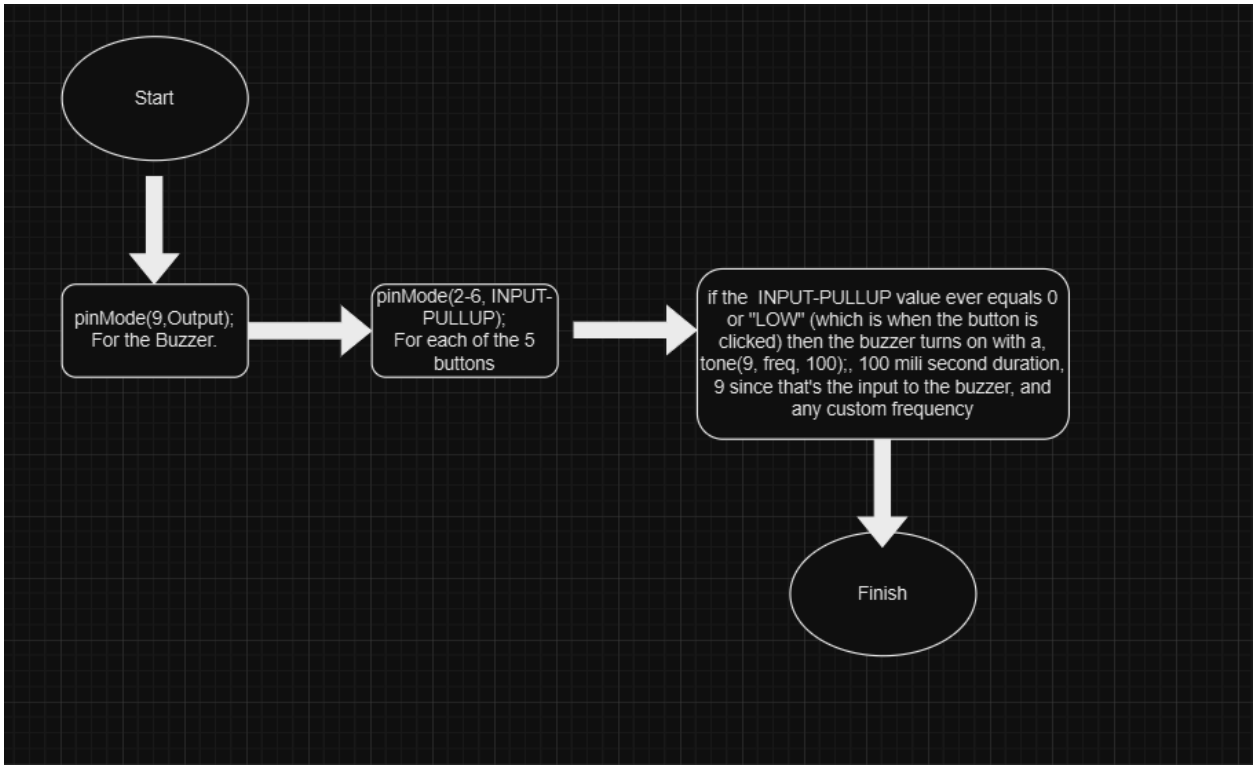
Requirements

The five buttons and the one buzzer is going to be set to a pinMode which is connected to the digital input values on the Arduino. Each of the five buttons is going to be connected to the buzzer and play the sounds of the keys C4, D4, E4, F4, and G4. We are going to do this by using a bunch of if statements and if a key is pressed it's going to play a frequency of one of the keys using the chart given in the lab procedure. For example, the input 6 button is going to play the buzzer at the frequency of 262 representing the C4 key. This will be done for the rest of the four buttons. The duration of each sound is going to stay constant at 100 milliseconds, the sound duration needs to be constant since in that way it's even and resembles a proper piano.

Wiring Diagram



Software Flowchart



Design Verification

https://youtube.com/shorts/BLLViLWOf8M - Video of Working Project

```
void setup() {
pinMode(9, OUTPUT);
pinMode(6, INPUT_PULLUP);
pinMode(5, INPUT_PULLUP);
pinMode(4, INPUT_PULLUP);
pinMode(3, INPUT_PULLUP);
pinMode(2, INPUT_PULLUP);
}
void loop() {
  // put your main code here, to run repeatedly:
if(digitalRead(6)==LOW)
{
tone(9,262,100);
}
if(digitalRead(5)==LOW)
{
tone(9,294,100);
}
if(digitalRead(4)==LOW)
{
tone(9,330,100);
}

if(digitalRead(3)==LOW)
{
tone(9,349,100);
}
if(digitalRead(2)==LOW)
{
tone(9,392,100);
}
}
```

Learnings

I learned more about grounding and how each piece of the bread board well as the bread board itself needs to be grounded. Also learned more about if statements and used them a lot in this project. Additionally, I learned the code to work a buzzer and how if it is clicked it is set to a low/0 value. So I got more familiar with setting the buzzer to 0.

Conclusion

I would say the project was a success since I was able to complete my goal of creating a functioning piano and everything went according to plan. One thing that I would do to better my project is to try to use fewer lines of code. Maybe add a for loop to cover all the buzzers in one big if statement rather then creating a specific if statement for each specific buzzer.